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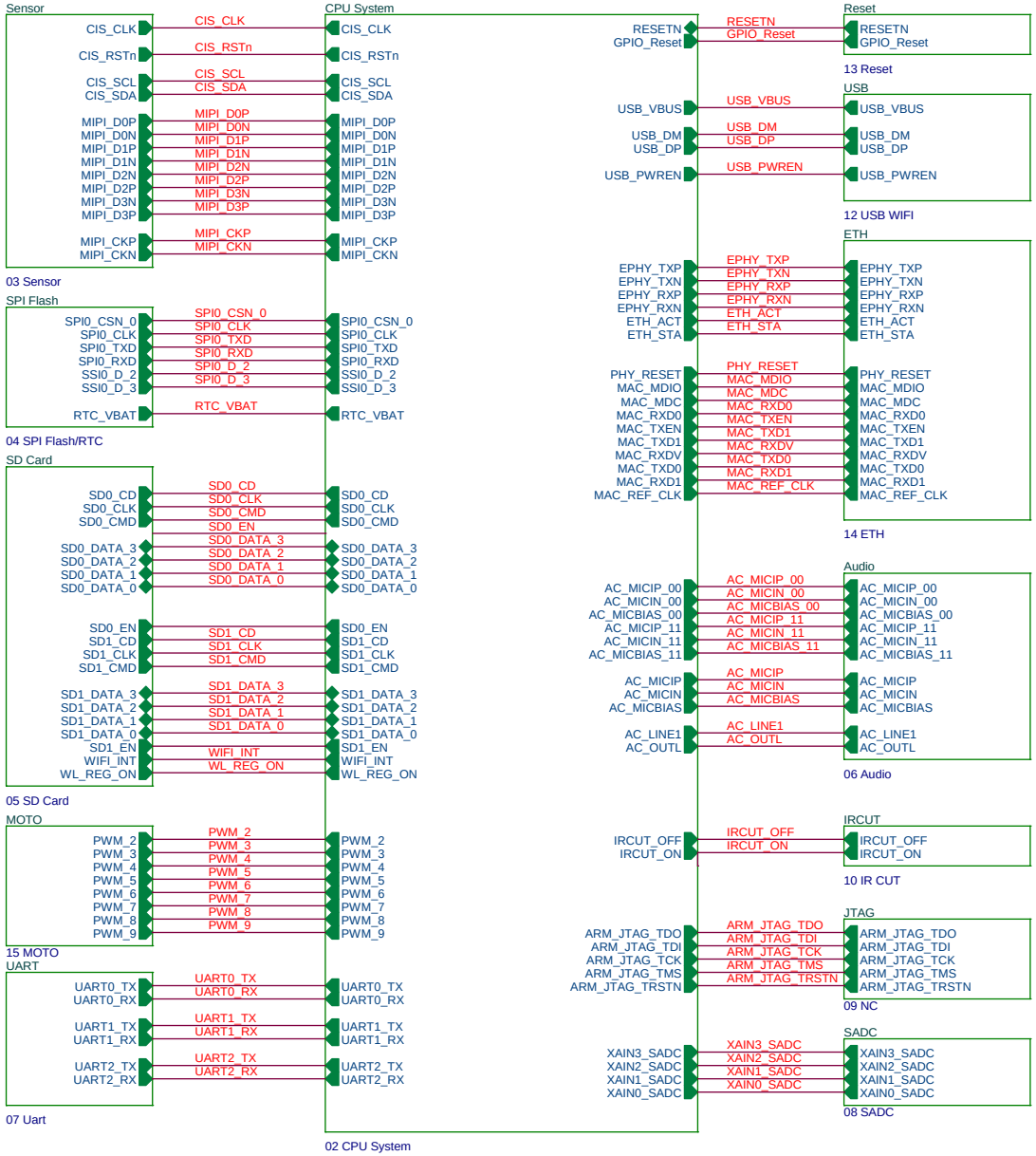

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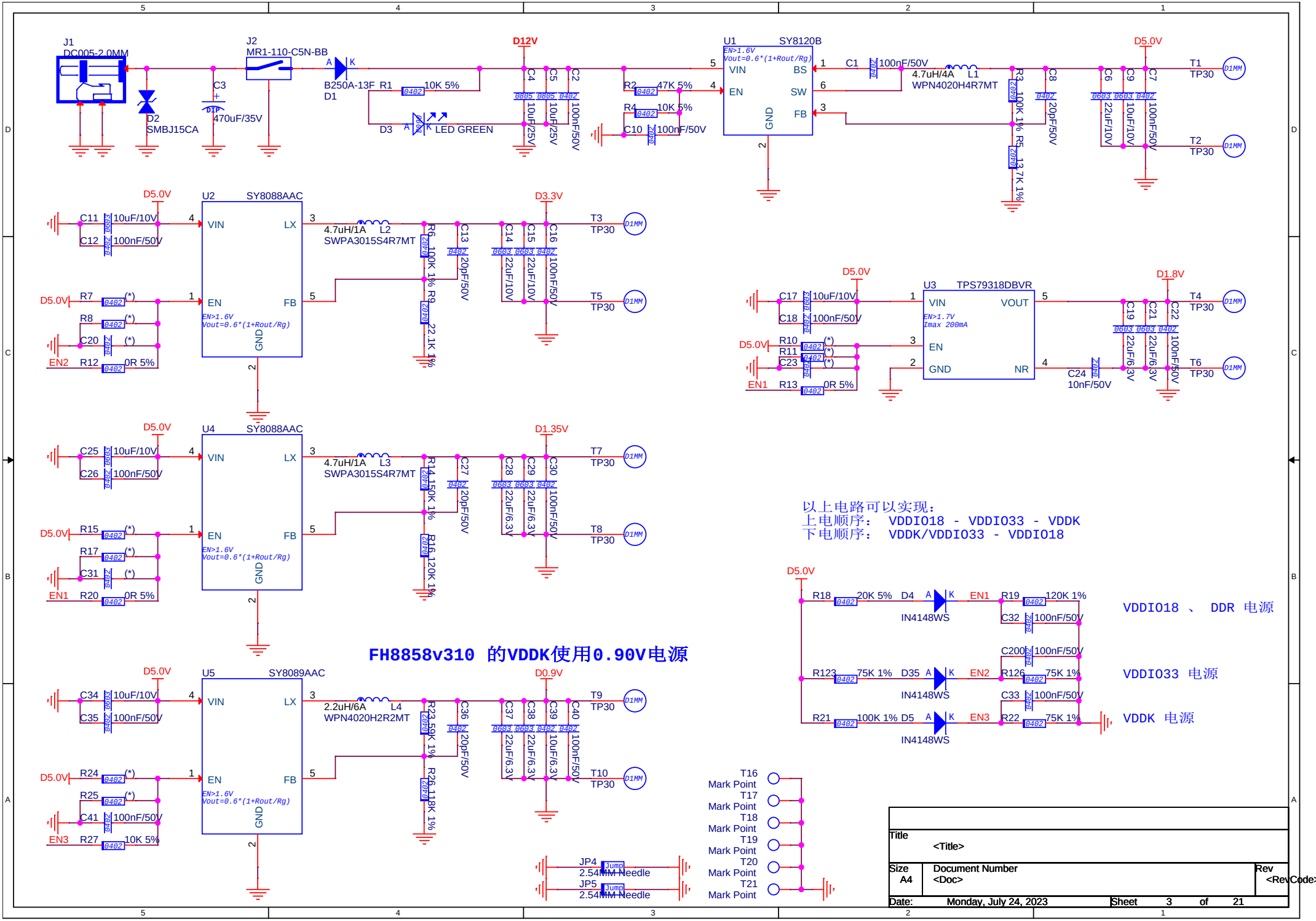
J1
DC005-2.0MM
J2
MR1-110-C5N-BB
D2
SMBJ15CA
C3
470uF/35V
B250A-13F
D1
R1
10K 5%
D3
LED GREEN
D12V
C4
10uF/25V
C5
10uF/25V
C2
100nF/50V
R2
47K 5%
R4
10K 5%
C10
100nF/50V
U1
SY8120B
EN>1.6V
Vout=0.6*(1+Rout/Rg)
VIN
EN
BS
SW
FB
GND
C1
100nF/50V
4.7uH/4A
L1
WPN4020H4R7MT
R3
100K 1%
R5
13.7K 1%
C8
20pF/50V
D5.0V
C6
22uF/10V
C9
10uF/10V
C7
100nF/50V
T1
TP30
T2
TP30
D1MM
U2
SY8088AAC
C11
10uF/10V
C12
100nF/50V
R7
R8
C20
R12
0R 5%
EN2
4.7uH/1A
L2
SWPA3015S4R7MT
R6
100K 1%
R9
22.1K 1%
C13
20pF/50V
C14
22uF/10V
C15
22uF/10V
C16
100nF/50V
D3.3V
T3
TP30
T5
TP30
U3
TPS79318DBVR
EN>1.7V
Imax 200mA
VIN
VOUT
EN
GND
NR
C17
10uF/10V
C18
100nF/50V
R10
R11
C23
R13
0R 5%
EN1
C24
10nF/50V
C19
22uF/6.3V
C21
22uF/6.3V
C22
100nF/50V
D1.8V
T4
TP30
T6
TP30
U4
SY8088AAC
C25
10uF/10V
C26
100nF/50V
R15
R17
C31
R20
0R 5%
EN1
4.7uH/1A
L3
SWPA3015S4R7MT
R14
150K 1%
R16
120K 1%
C27
20pF/50V
C28
22uF/6.3V
C29
22uF/6.3V
C30
100nF/50V
D1.35V
T7
TP30
T8
TP30
以上电路可以实现：
上电顺序： VDDIO18 - VDDIO33 - VDDK
下电顺序： VDDK/VDDIO33 - VDDIO18
R18
20K 5%
D4
IN4148WS
EN1
R19
120K 1%
C32
100nF/50V
VDDIO18 、 DDR 电源
R123
75K 1%
D35
IN4148WS
EN2
R126
75K 1%
C200
100nF/50V
C33
100nF/50V
VDDIO33 电源
R21
100K 1%
D5
IN4148WS
EN3
R22
75K 1%
VDDK 电源
FH8858v310 的VDDK使用0.90V电源
U5
SY8089AAC
C34
10uF/10V
C35
100nF/50V
R24
R25
C41
100nF/50V
R27
10K 5%
EN3
2.2uH/6A
L4
WPN4020H2R2MT
R23
59K 1%
R26
118K 1%
C36
20pF/50V
C37
22uF/6.3V
C38
22uF/6.3V
C39
10uF/6.3V
C40
100nF/50V
D0.9V
T9
TP30
T10
TP30
JP4
JP5
2.54MM Needle
T16
Mark Point
T17
Mark Point
T18
Mark Point
T19
Mark Point
T20
Mark Point
T21
Mark Point
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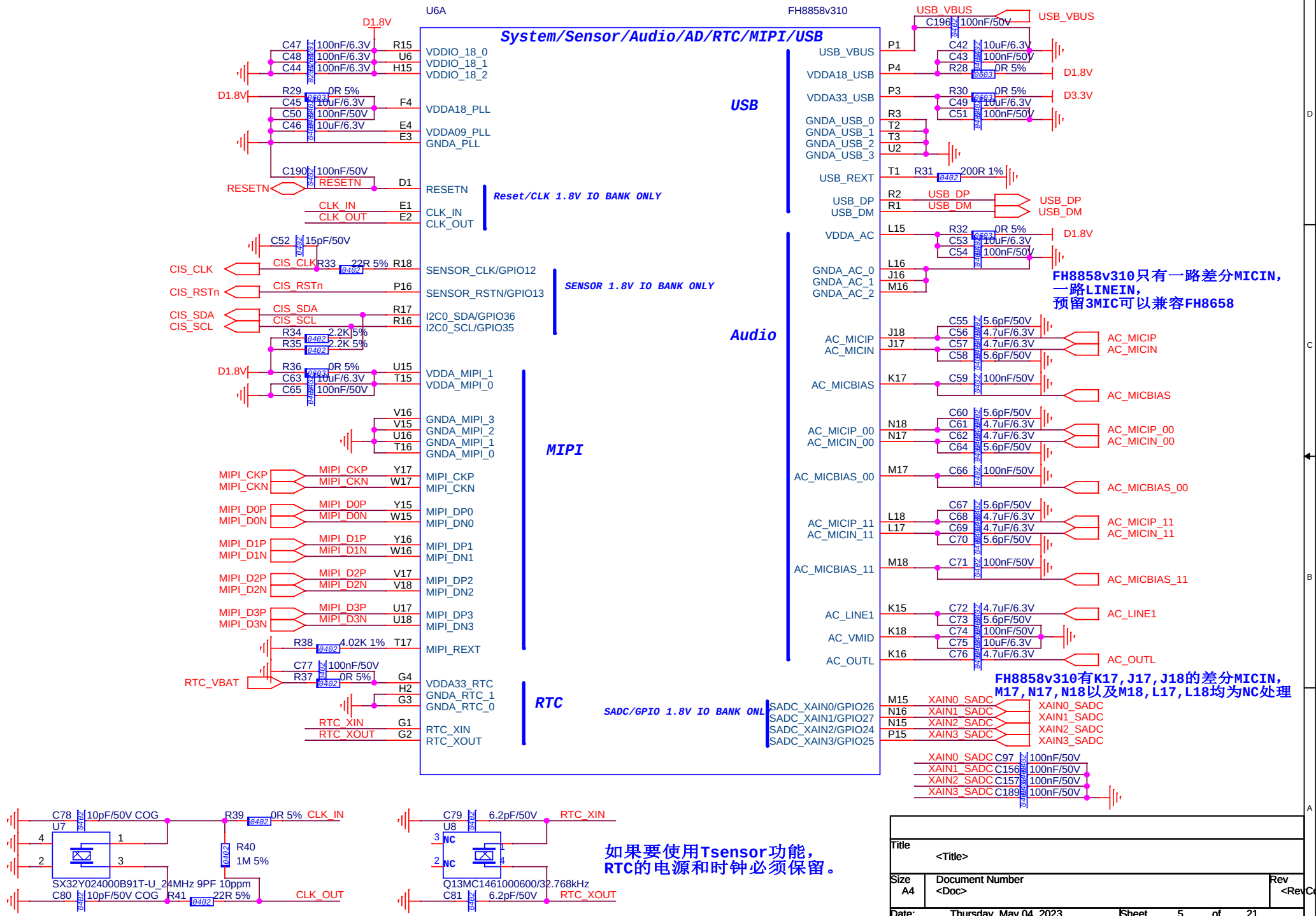
U6A
FH8858v310
System/Sensor/Audio/AD/RTC/MIPI/USB
USB
Reset/CLK 1.8V IO BANK ONLY
SENSOR 1.8V IO BANK ONLY
Audio
MIPI
RTC
SADC/GPIO 1.8V IO BANK ONL
VDDIO_18_0
VDDIO_18_1
VDDIO_18_2
VDDA18_PLL
VDDA09_PLL
GNDA_PLL
RESETN
CLK_IN
CLK_OUT
SENSOR_CLK/GPIO12
SENSOR_RSTN/GPIO13
I2C0_SDA/GPIO36
I2C0_SCL/GPIO35
VDDA_MIPI_1
VDDA_MIPI_0
GNDA_MIPI_3
GNDA_MIPI_2
GNDA_MIPI_1
GNDA_MIPI_0
MIPI_CKP
MIPI_CKN
MIPI_DP0
MIPI_DN0
MIPI_DP1
MIPI_DN1
MIPI_DP2
MIPI_DN2
MIPI_DP3
MIPI_DN3
MIPI_REXT
VDDA33_RTC
GNDA_RTC_1
GNDA_RTC_0
RTC_XIN
RTC_XOUT
SADC_XAIN0/GPIO26
SADC_XAIN1/GPIO27
SADC_XAIN2/GPIO24
SADC_XAIN3/GPIO25
USB_VBUS
VDDA18_USB
VDDA33_USB
GNDA_USB_0
GNDA_USB_1
GNDA_USB_2
GNDA_USB_3
USB_REXT
USB_DP
USB_DM
VDDA_AC
GNDA_AC_0
GNDA_AC_1
GNDA_AC_2
AC_MICIP
AC_MICIN
AC_MICBIAS
AC_MICIP_00
AC_MICIN_00
AC_MICBIAS_00
AC_MICIP_11
AC_MICIN_11
AC_MICBIAS_11
AC_LINE1
AC_VMID
AC_OUTL
XAIN0_SADC
XAIN1_SADC
XAIN2_SADC
XAIN3_SADC
D1.8V
D3.3V
FH8858v310只有一路差分MICIN,
一路LINEIN,
预留3MIC可以兼容FH8658
FH8858v310有K17,J17,J18的差分MICIN,
M17,N17,N18以及M18,L17,L18均为NC处理
SX32Y024000B91T-U_24MHz 9PF 10ppm
Q13MC1461000600/32.768kHz
如果要使用Tsensor功能,
RTC的电源和时钟必须保留。
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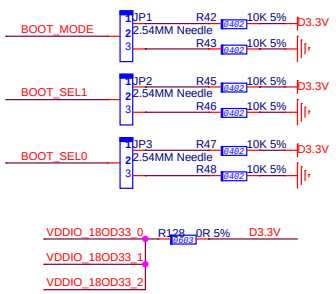
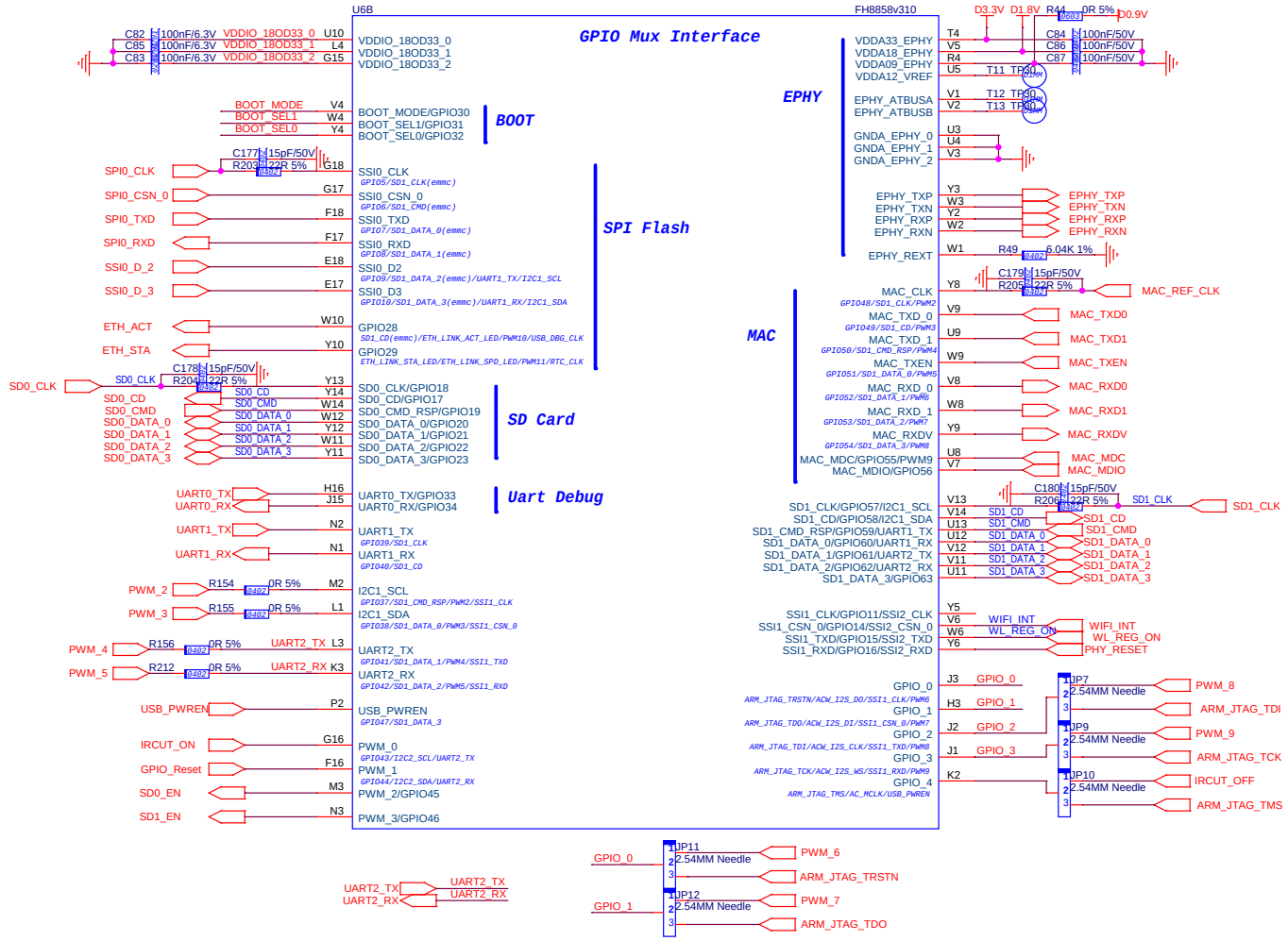
BOOT_MODE GPIO30	BOOT_SEL1 GPIO31	BOOY_SEL0 GPIO32	MODE
0	X	X	Uart Download
1	1	1	Nor Flash
1	1	0	SD0 Emmc
1	0	1	Nand Flash
1	0	0	SD1 Emmc





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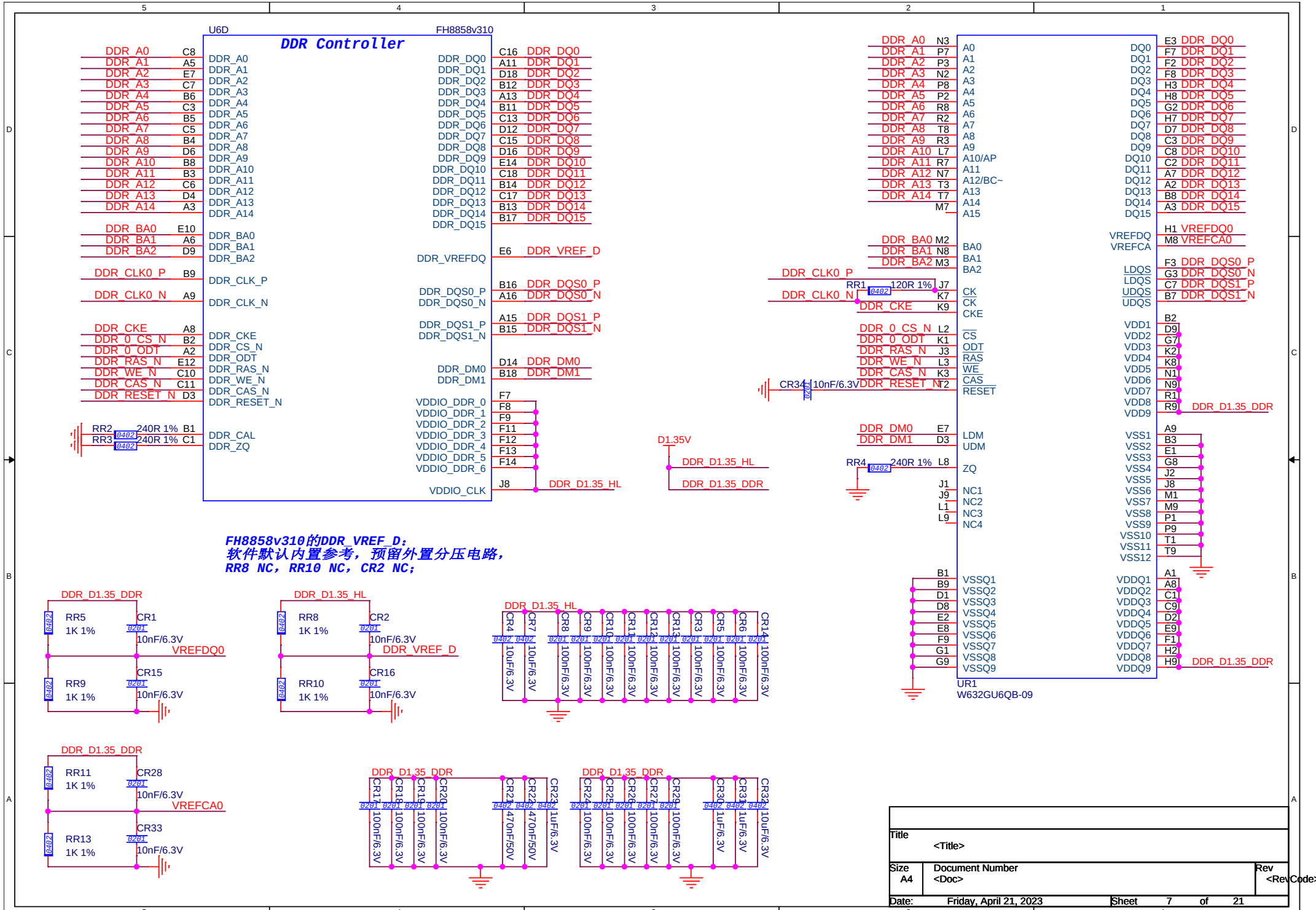
U6D
FH8858v310
DDR Controller
DDR_A0 C8
DDR_A1 A5
DDR_A2 E7
DDR_A3 C7
DDR_A4 B6
DDR_A5 C3
DDR_A6 B5
DDR_A7 C5
DDR_A8 B4
DDR_A9 D6
DDR_A10 B8
DDR_A11 B3
DDR_A12 C6
DDR_A13 D4
DDR_A14 A3
DDR_BA0 E10
DDR_BA1 A6
DDR_BA2 D9
DDR_CLK0_P B9
DDR_CLK0_N A9
DDR_CKE A8
DDR_0_CS_N B2
DDR_0_ODT A2
DDR_RAS_N E12
DDR_WE_N C10
DDR_CAS_N C11
DDR_RESET_N D3
RR2 0402 240R 1% B1
RR3 0402 240R 1% C1
DDR_CAL
DDR_ZQ
DDR_DQ0 C16
DDR_DQ1 A11
DDR_DQ2 D18
DDR_DQ3 B12
DDR_DQ4 A13
DDR_DQ5 B11
DDR_DQ6 C13
DDR_DQ7 D12
DDR_DQ8 C15
DDR_DQ9 D16
DDR_DQ10 E14
DDR_DQ11 C18
DDR_DQ12 B14
DDR_DQ13 C17
DDR_DQ14 B13
DDR_DQ15 B17
DDR_VREFDQ E6 DDR_VREF_D
DDR_DQS0_P B16
DDR_DQS0_N A16
DDR_DQS1_P A15
DDR_DQS1_N B15
DDR_DM0 D14
DDR_DM1 B18
VDDIO_DDR_0 F7
VDDIO_DDR_1 F8
VDDIO_DDR_2 F9
VDDIO_DDR_3 F11
VDDIO_DDR_4 F12
VDDIO_DDR_5 F13
VDDIO_DDR_6 F14
VDDIO_CLK J8 DDR_D1.35_HL
D1.35V
DDR_D1.35_HL
DDR_D1.35_DDR
DDR_A0 N3
DDR_A1 P7
DDR_A2 P3
DDR_A3 N2
DDR_A4 P8
DDR_A5 P2
DDR_A6 R8
DDR_A7 R2
DDR_A8 T8
DDR_A9 R3
DDR_A10 L7
DDR_A11 R7
DDR_A12 N7
DDR_A13 T3
DDR_A14 T7
M7
DDR_BA0 M2
DDR_BA1 N8
DDR_BA2 M3
DDR_CLK0_P
RR1 0402 120R 1% J7
DDR_CLK0_N K7
DDR_CKE K9
DDR_0_CS_N L2
DDR_0_ODT K1
DDR_RAS_N J3
DDR_WE_N L3
DDR_CAS_N K3
CR34 0402 10nF/6.3V
DDR_RESET_N T2
DDR_DM0 E7
DDR_DM1 D3
RR4 0402 240R 1% L8
J1
J9
L1
L9
NC1
NC2
NC3
NC4
B1
B9
D1
D8
E2
E8
F9
G1
G9
VSSQ1
VSSQ2
VSSQ3
VSSQ4
VSSQ5
VSSQ6
VSSQ7
VSSQ8
VSSQ9
UR1
W632GU6QB-09
DQ0 E3 DDR_DQ0
DQ1 F7 DDR_DQ1
DQ2 F2 DDR_DQ2
DQ3 F8 DDR_DQ3
DQ4 H3 DDR_DQ4
DQ5 H8 DDR_DQ5
DQ6 G2 DDR_DQ6
DQ7 H7 DDR_DQ7
DQ8 D7 DDR_DQ8
DQ9 C3 DDR_DQ9
DQ10 C8 DDR_DQ10
DQ11 C2 DDR_DQ11
DQ12 A7 DDR_DQ12
DQ13 A2 DDR_DQ13
DQ14 B8 DDR_DQ14
DQ15 A3 DDR_DQ15
VREFDQ H1 VREFDQ0
VREFCA M8 VREFCA0
LDQS F3 DDR_DQS0_P
LDQS G3 DDR_DQS0_N
UDQS C7 DDR_DQS1_P
UDQS B7 DDR_DQS1_N
VDD1 B2
VDD2 D9
VDD3 G7
VDD4 K2
VDD5 K8
VDD6 N1
VDD7 N9
VDD8 R1
VDD9 R9 DDR_D1.35_DDR
VSS1 A9
VSS2 B3
VSS3 E1
VSS4 G8
VSS5 J2
VSS6 J8
VSS7 M1
VSS8 M9
VSS9 P1
VSS10 P9
VSS11 T1
VSS12 T9
VDDQ1 A1
VDDQ2 A8
VDDQ3 C1
VDDQ4 C9
VDDQ5 D2
VDDQ6 E9
VDDQ7 F1
VDDQ8 H2
VDDQ9 H9 DDR_D1.35_DDR
FH8858v310的DDR_VREF_D;
软件默认内置参考，预留外置分压电路，
RR8 NC，RR10 NC，CR2 NC;
DDR_D1.35_DDR
RR5 0402 1K 1%
CR1 0201 10nF/6.3V
VREFDQ0
RR9 0402 1K 1%
CR15 0201 10nF/6.3V
DDR_D1.35_HL
RR8 0402 1K 1%
CR2 0201 10nF/6.3V
DDR_VREF_D
RR10 0402 1K 1%
CR16 0201 10nF/6.3V
DDR_D1.35_HL
CR4 0402 10uF/6.3V
CR7 0402 10uF/6.3V
CR8 0201 100nF/6.3V
CR9 0201 100nF/6.3V
CR10 0201 100nF/6.3V
CR11 0201 100nF/6.3V
CR12 0201 100nF/6.3V
CR13 0201 100nF/6.3V
CR3 0201 100nF/6.3V
CR5 0201 100nF/6.3V
CR6 0201 100nF/6.3V
CR14 0201 100nF/6.3V
DDR_D1.35_DDR
RR11 0402 1K 1%
CR28 0201 10nF/6.3V
VREFCA0
RR13 0402 1K 1%
CR33 0201 10nF/6.3V
DDR_D1.35_DDR
CR17 0201 100nF/6.3V
CR18 0201 100nF/6.3V
CR19 0201 100nF/6.3V
CR20 0201 100nF/6.3V
CR21 0402 470nF/50V
CR22 0402 470nF/50V
CR23 0402 1uF/6.3V
DDR_D1.35_DDR
CR24 0201 100nF/6.3V
CR25 0201 100nF/6.3V
CR26 0201 100nF/6.3V
CR27 0201 100nF/6.3V
CR29 0201 100nF/6.3V
CR30 0402 1uF/6.3V
CR31 0402 1uF/6.3V
CR32 0402 10uF/6.3V
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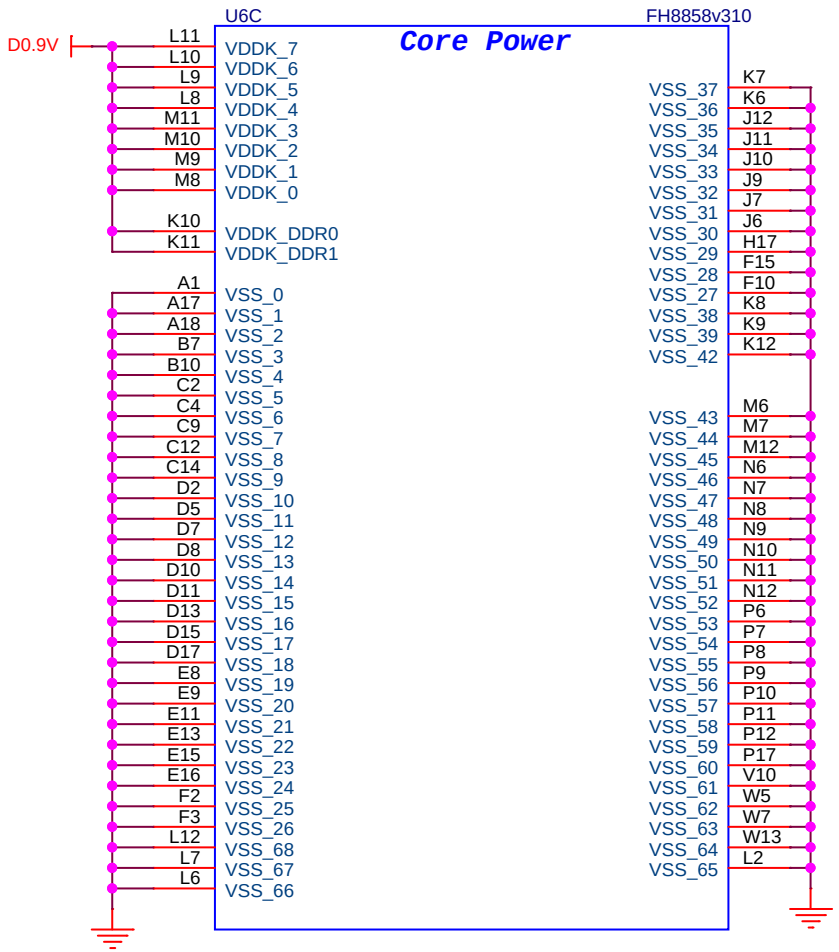

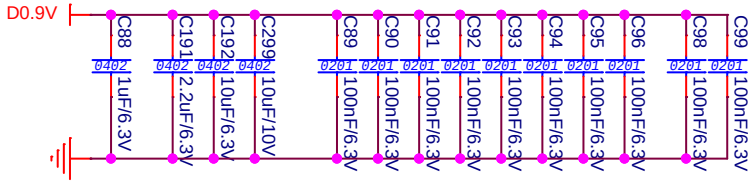

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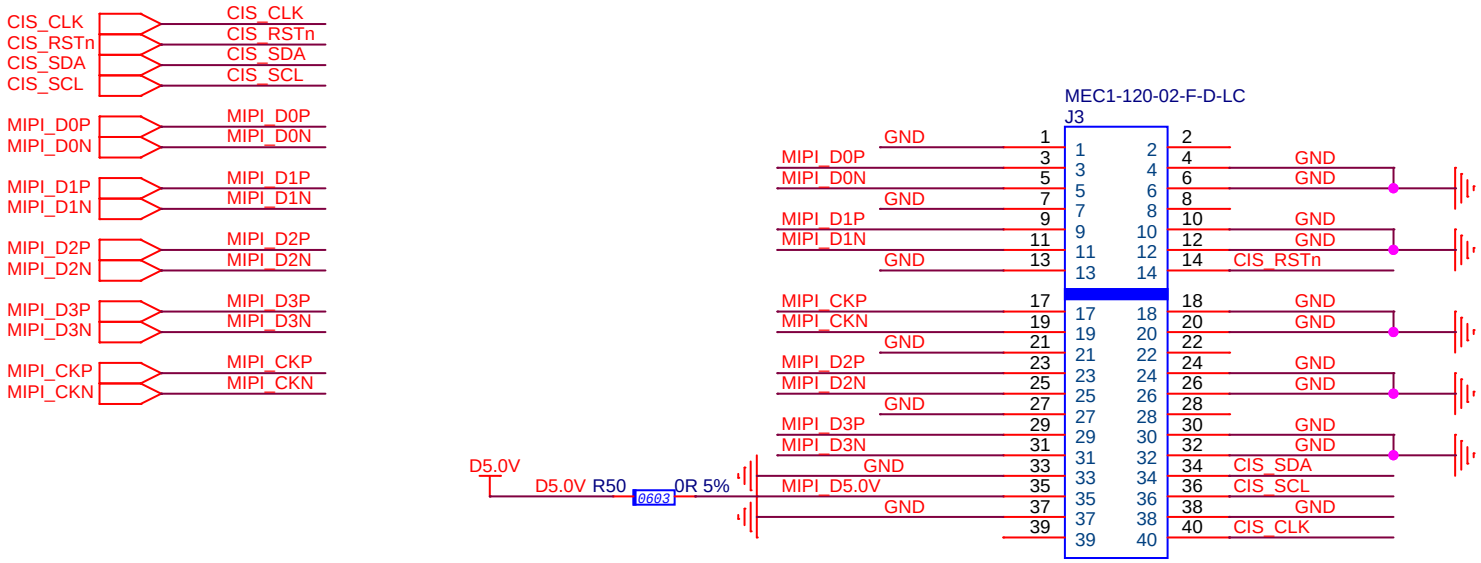

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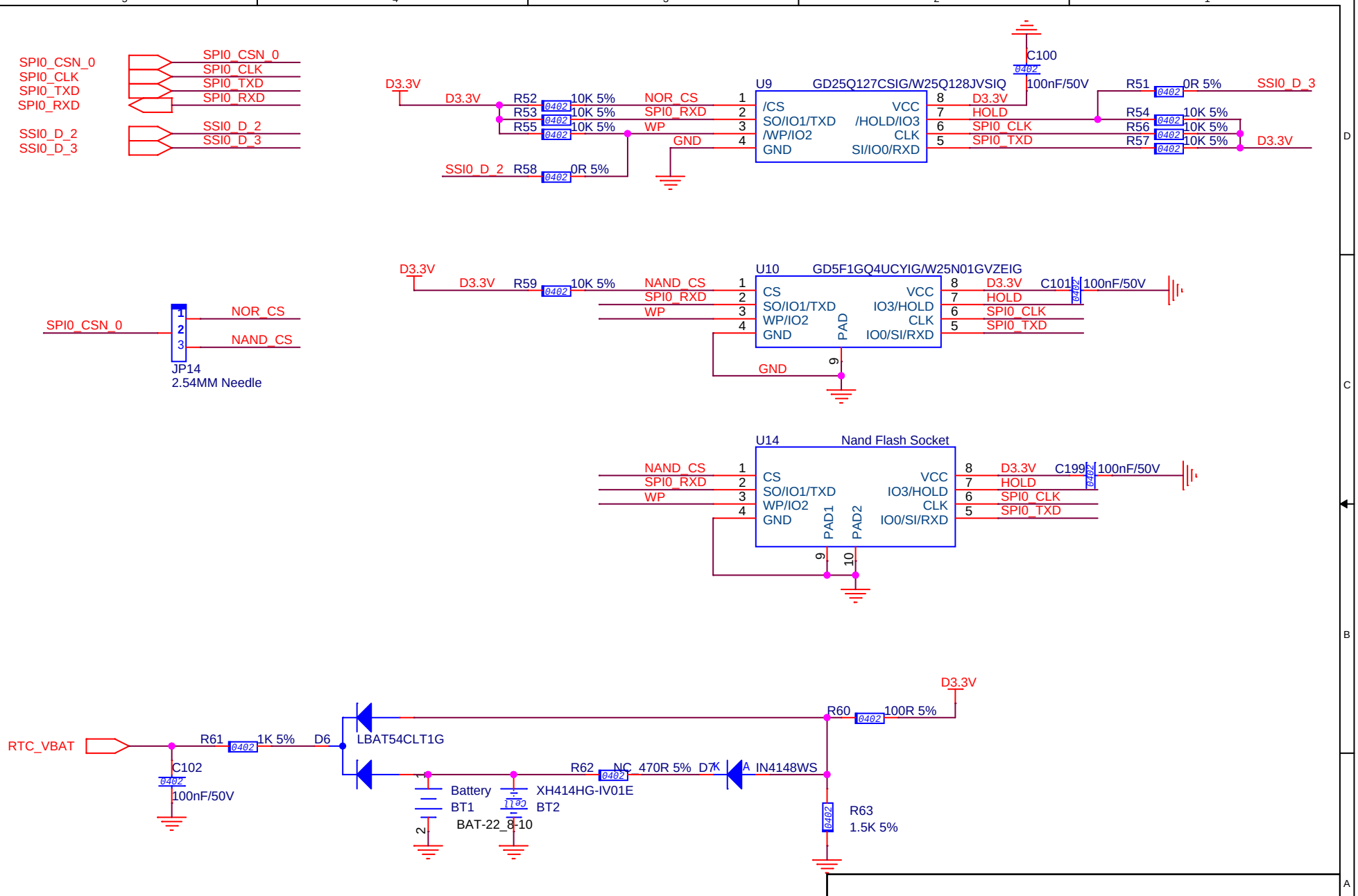
SPI0_CSN_0
SPI0_CLK
SPI0_TXD
SPI0_RXD
SSI0_D_2
SSI0_D_3
D3.3V
R52 10K 5%
R53 10K 5%
R55 10K 5%
R58 0R 5%
NOR_CS
SPI0_RXD
WP
GND
U9 GD25Q127CSIG/W25Q128JVSIQ
/CS
SO/IO1/TXD
/WP/IO2
GND
VCC
/HOLD/IO3
CLK
SI/IO0/RXD
HOLD
C100 100nF/50V
R51 0R 5%
R54 10K 5%
R56 10K 5%
R57 10K 5%
JP14
2.54MM Needle
NAND_CS
U10 GD5F1GQ4UCYIG/W25N01GVZEIG
CS
SO/IO1/TXD
WP/IO2
GND
PAD
VCC
IO3/HOLD
CLK
IO0/SI/RXD
C101 100nF/50V
U14 Nand Flash Socket
PAD1
PAD2
C199 100nF/50V
RTC_VBAT
R61 1K 5%
D6
LBAT54CLT1G
C102 100nF/50V
R60 100R 5%
R62 NC 470R 5%
D7K
A IN4148WS
Battery
BT1
BAT-22_8-10
XH414HG-IV01E
BT2
R63 1.5K 5%
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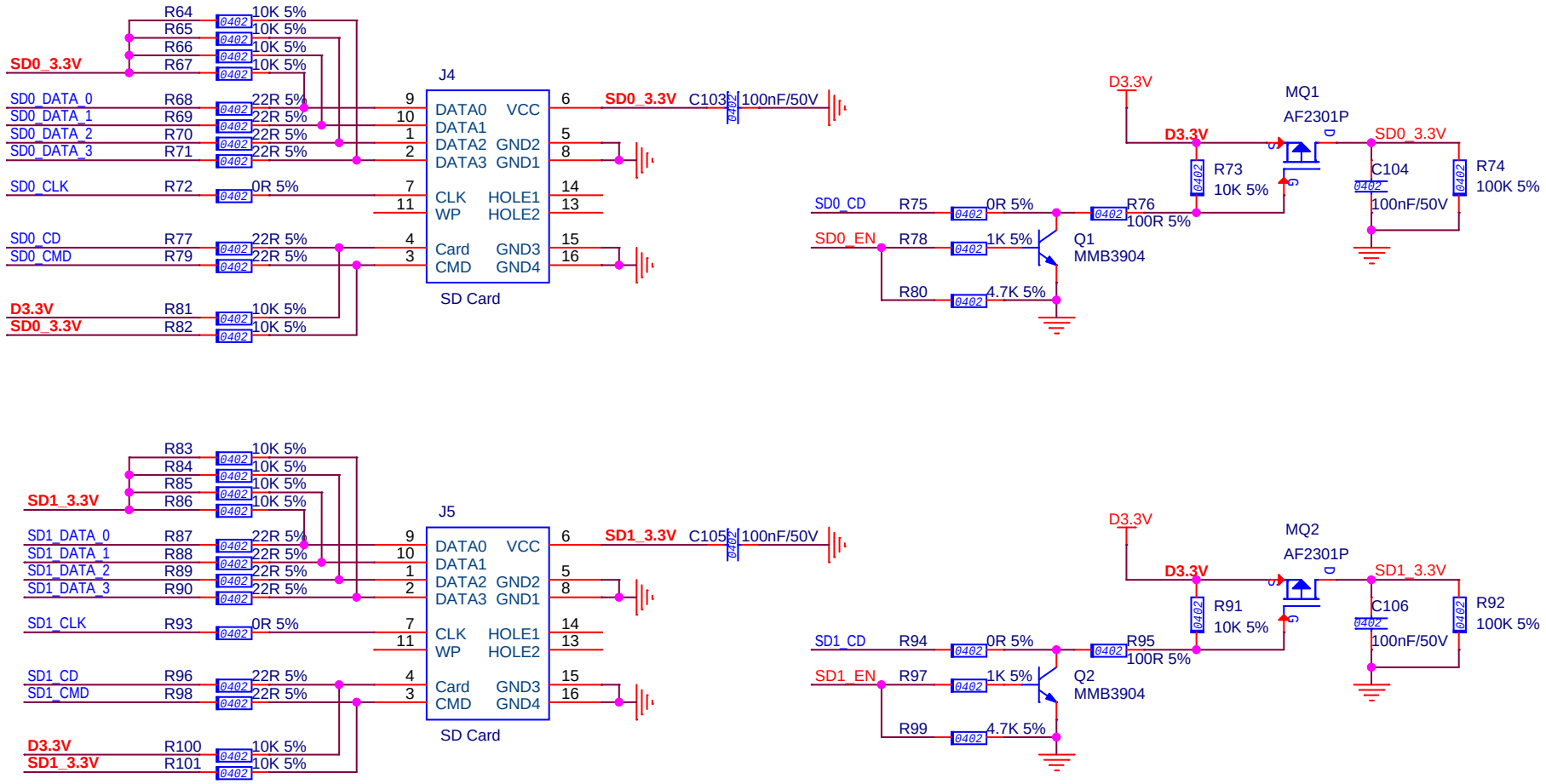

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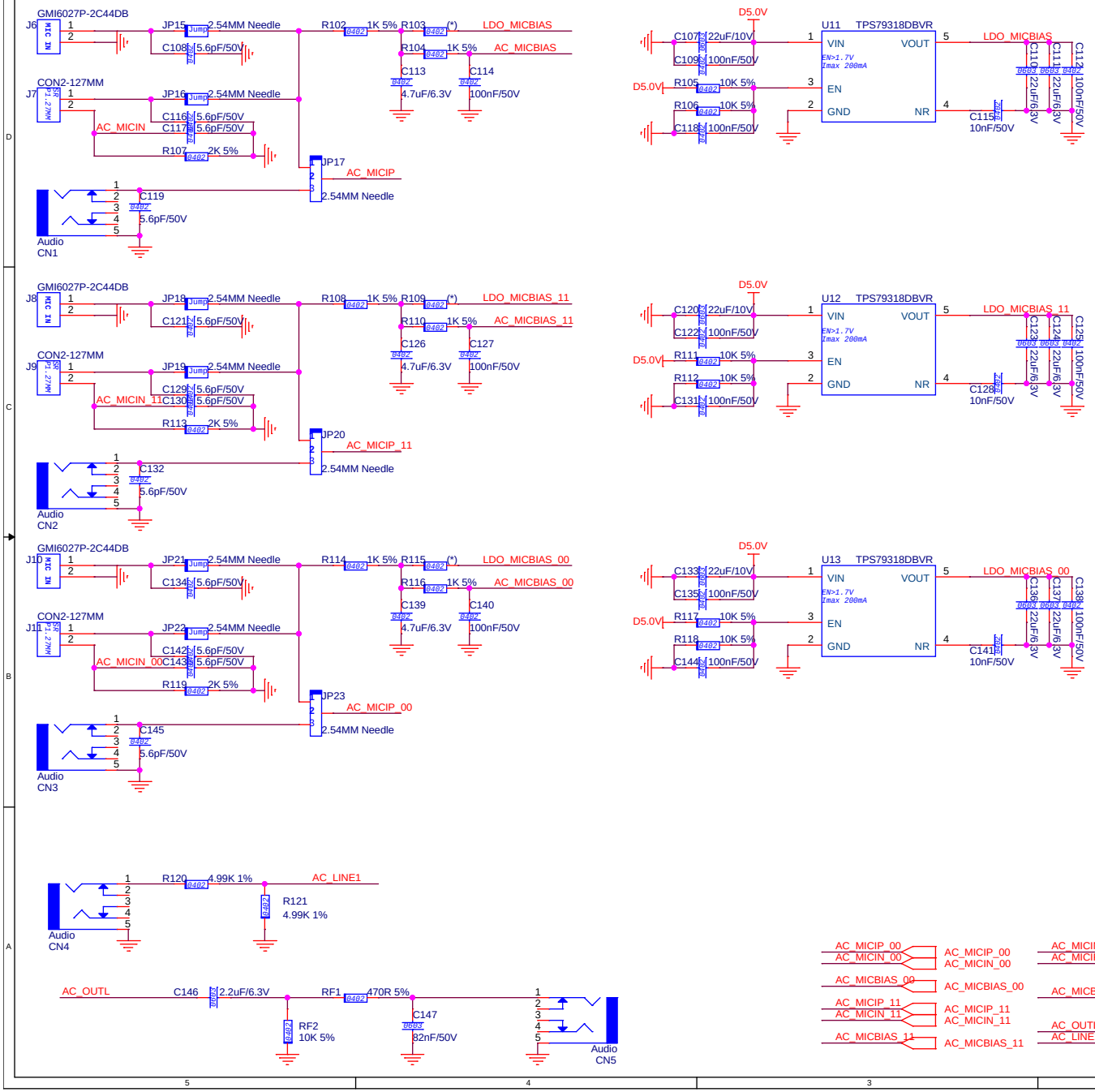

FH8858v310只有一路差分MICIN，一路LINEIN，预留3MIC可以兼容FH8658

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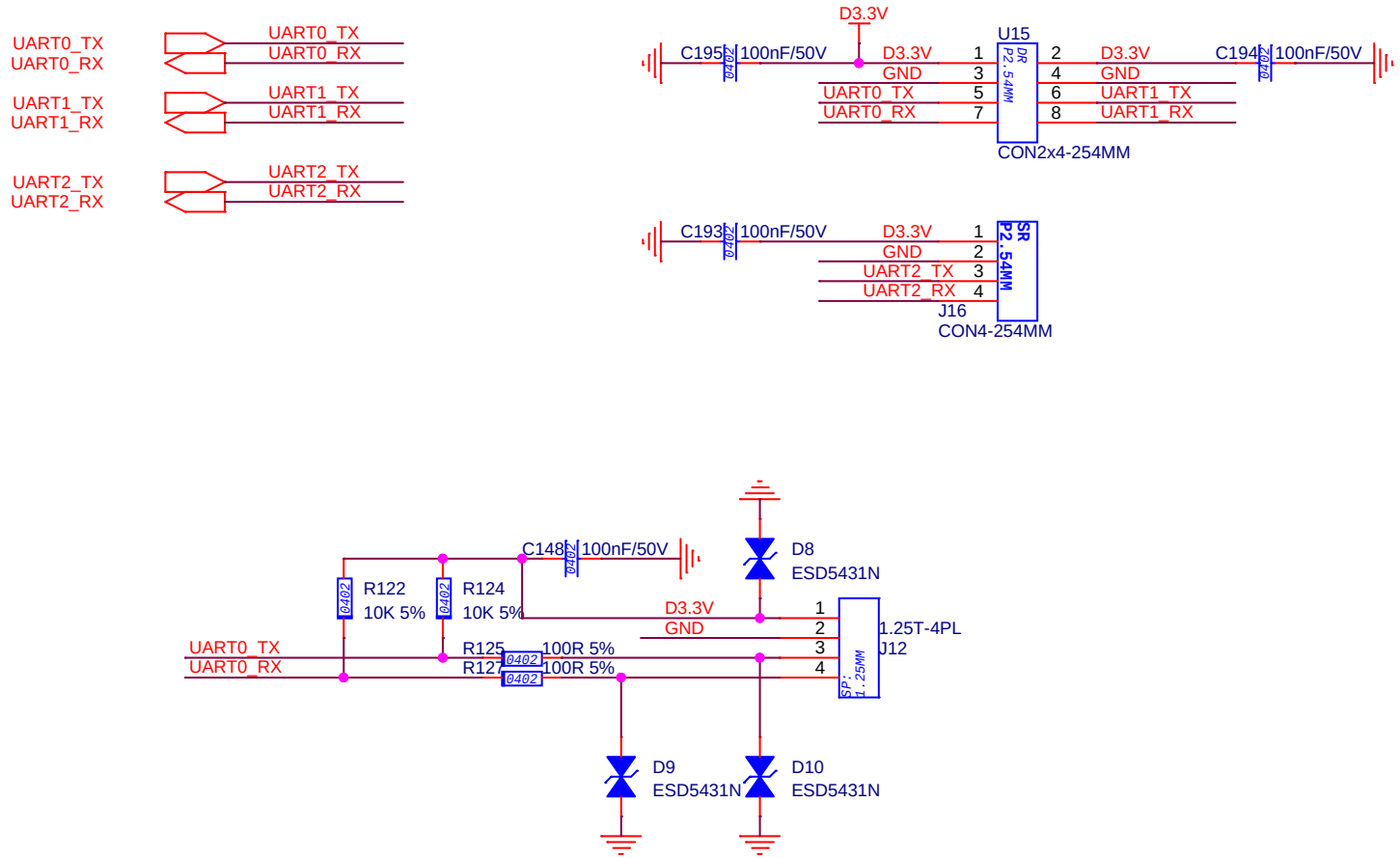

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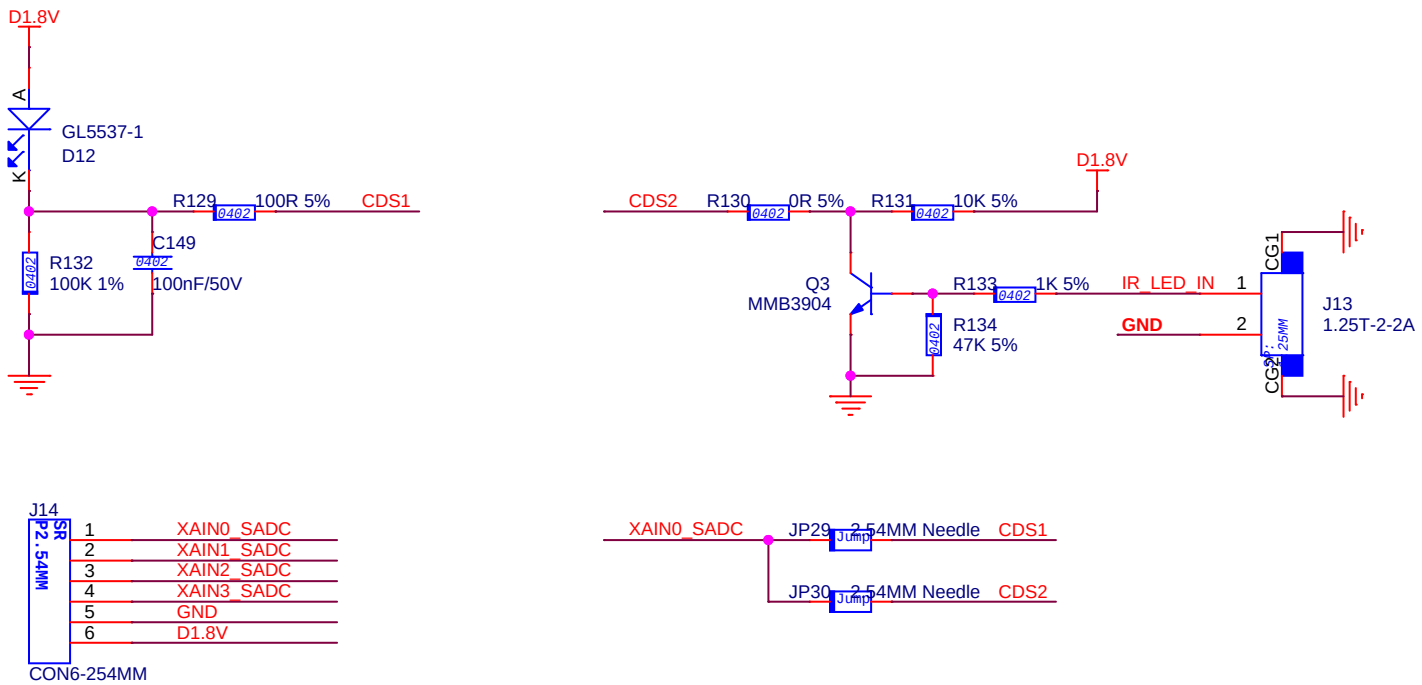

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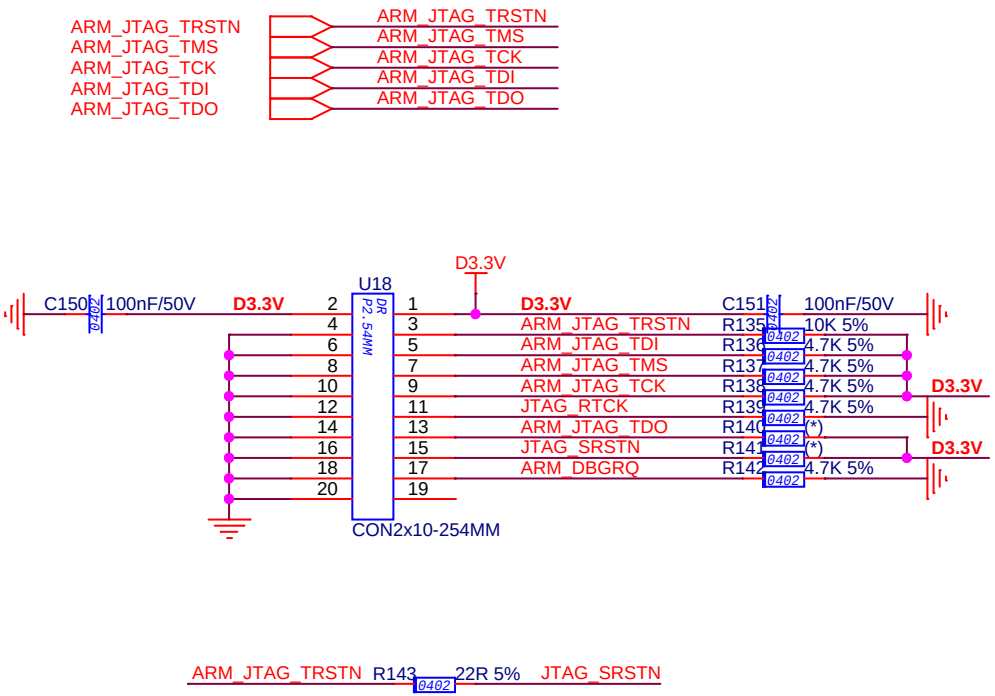

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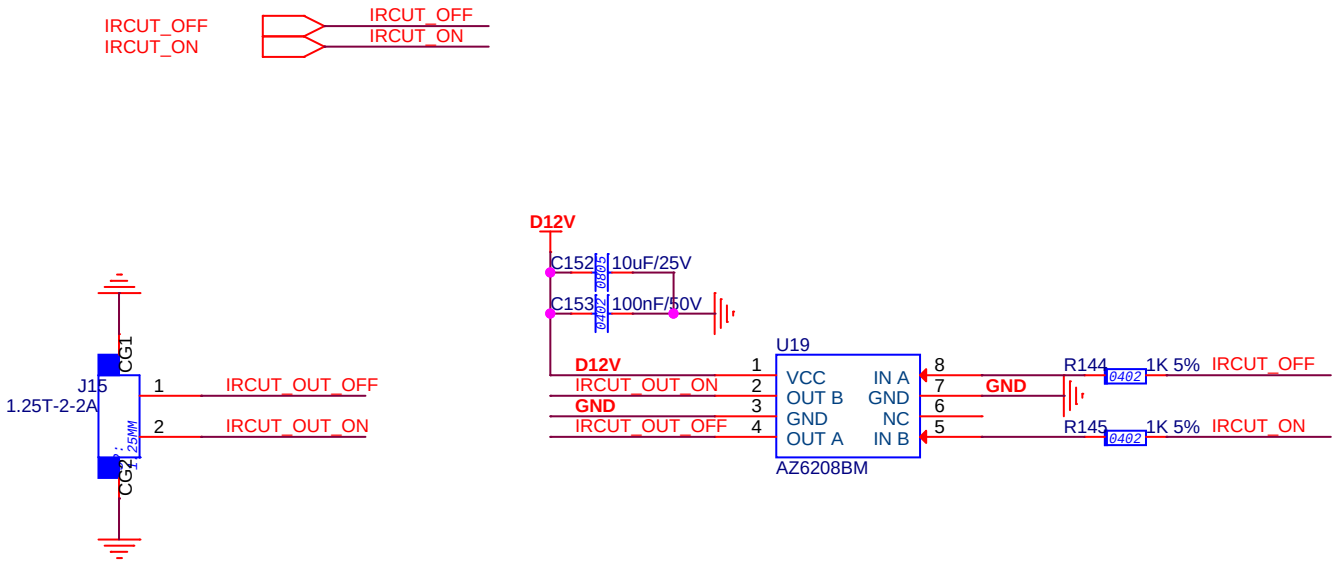

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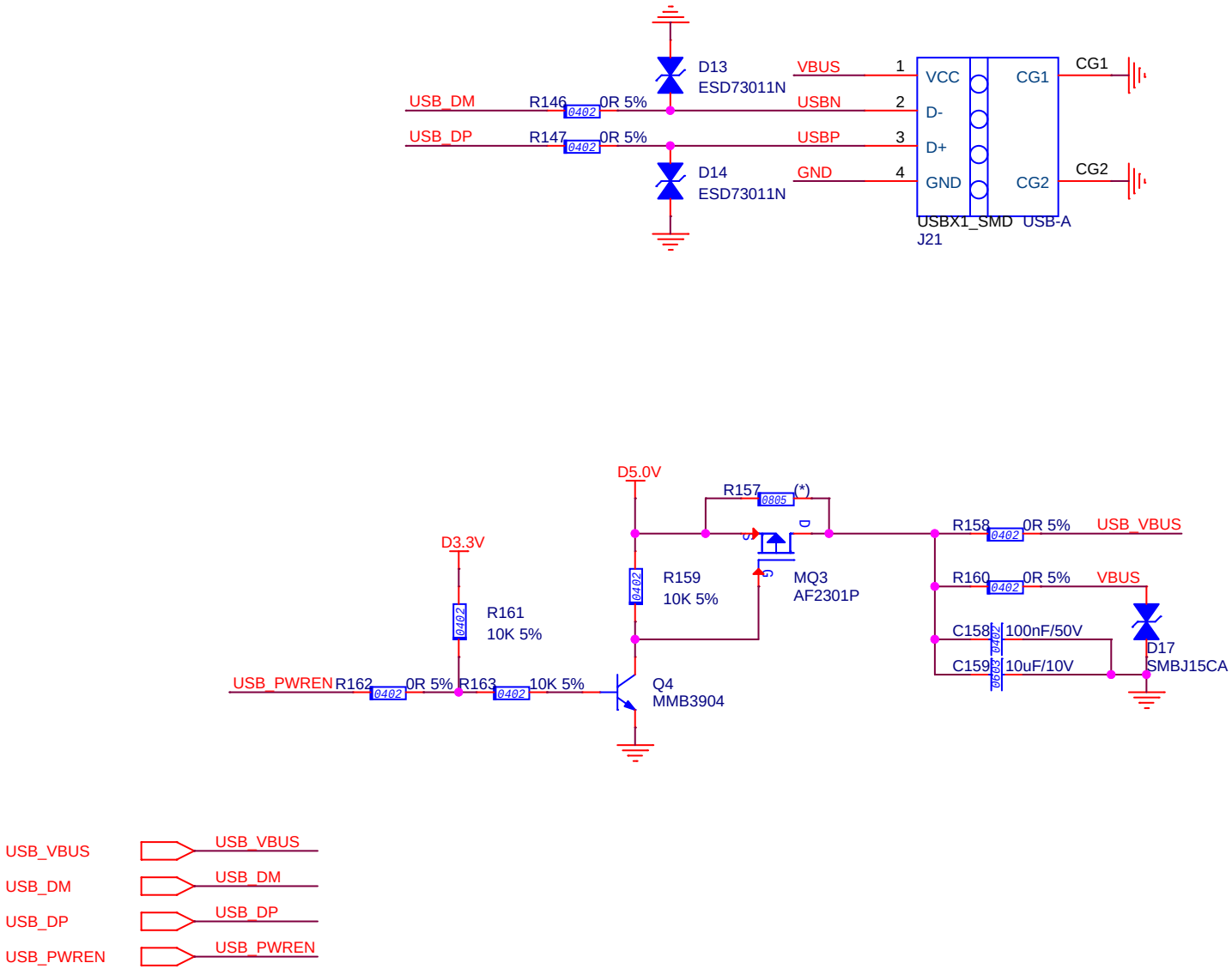

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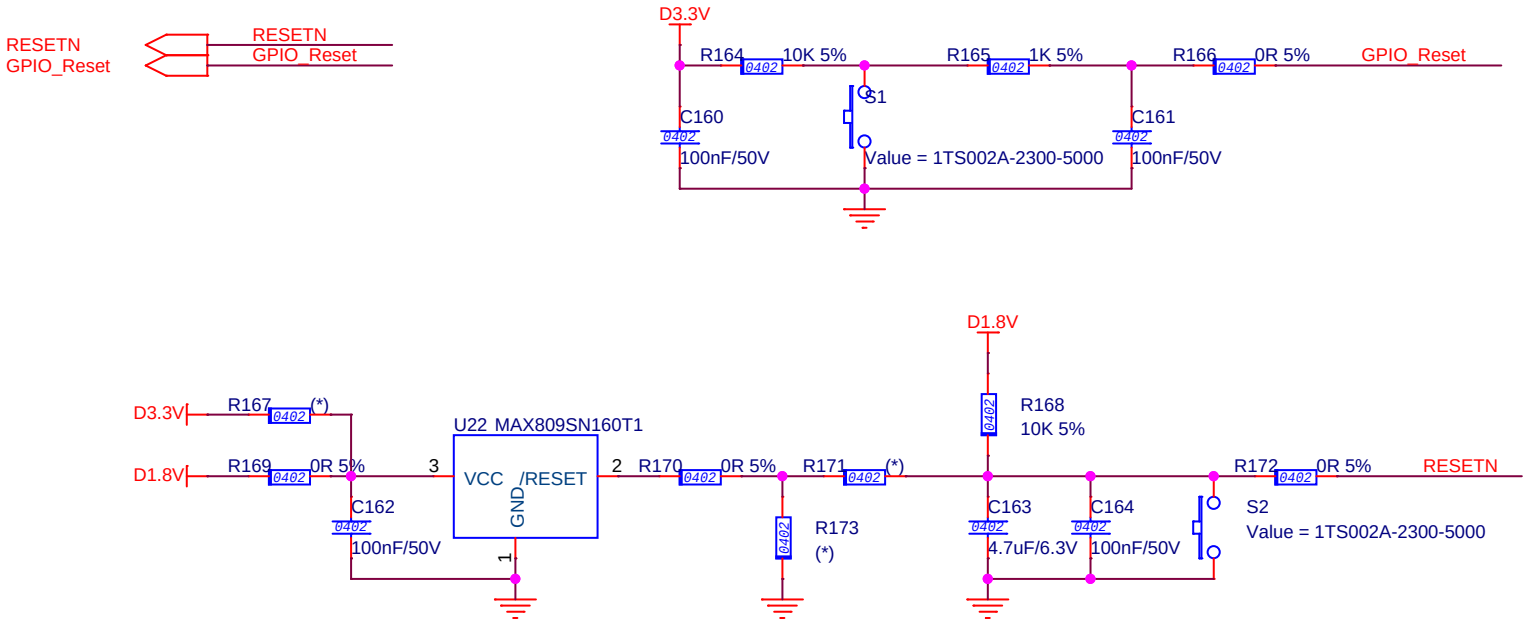

FH8858v310可以使用RC复位电路完成复位功能；

预留复位芯片，可以兼容FH885xv200/FH865x系列，推荐 th=2.63V
可以用3.3v复位芯片分压得到1.8v复位信号，分压电阻不能太大，推荐使用参考值

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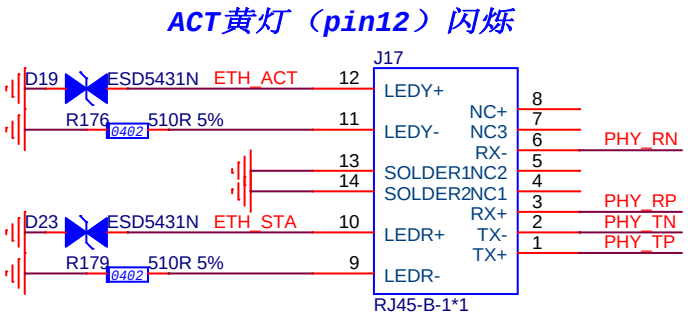
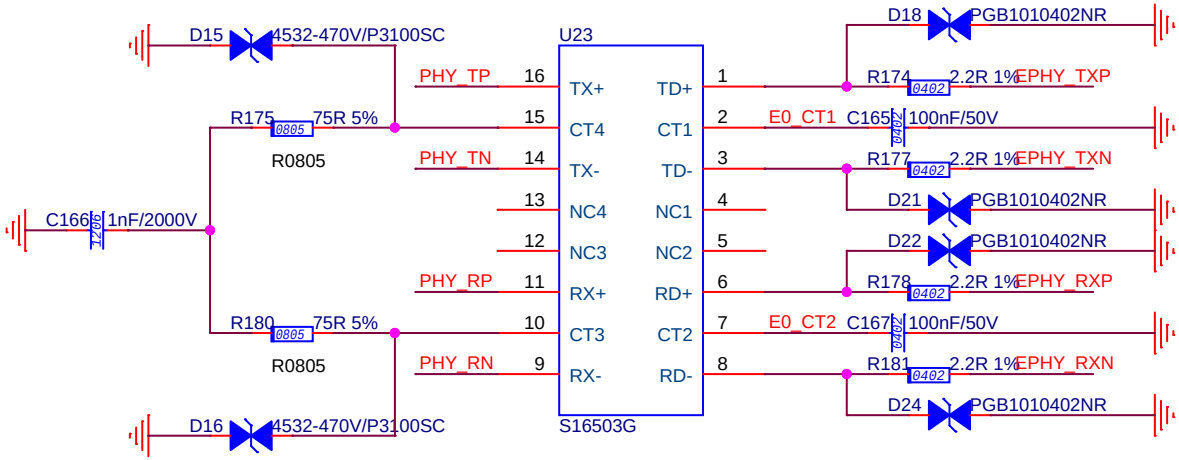
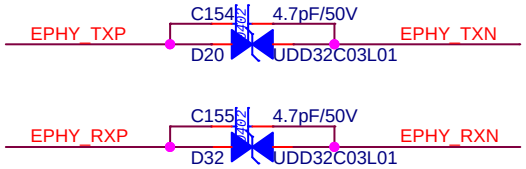


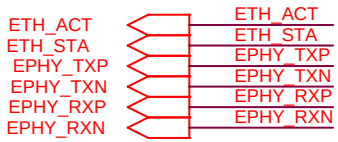


初级　　次级

TX/RX 上的 4 个串联电阻 2.2R ，可以保证信号完整性；
D15/D16 使用放电管，作 ESD 防护，可只上件任意一个；
R175/R180 使用 0805 封装，以防被击穿，可合二为一个 75 欧电阻



差 分电容 4.7pF 可以保证信号完整性，以 及抑制 EMC ；
差 分 TVS 抑制 次级 ESD ；



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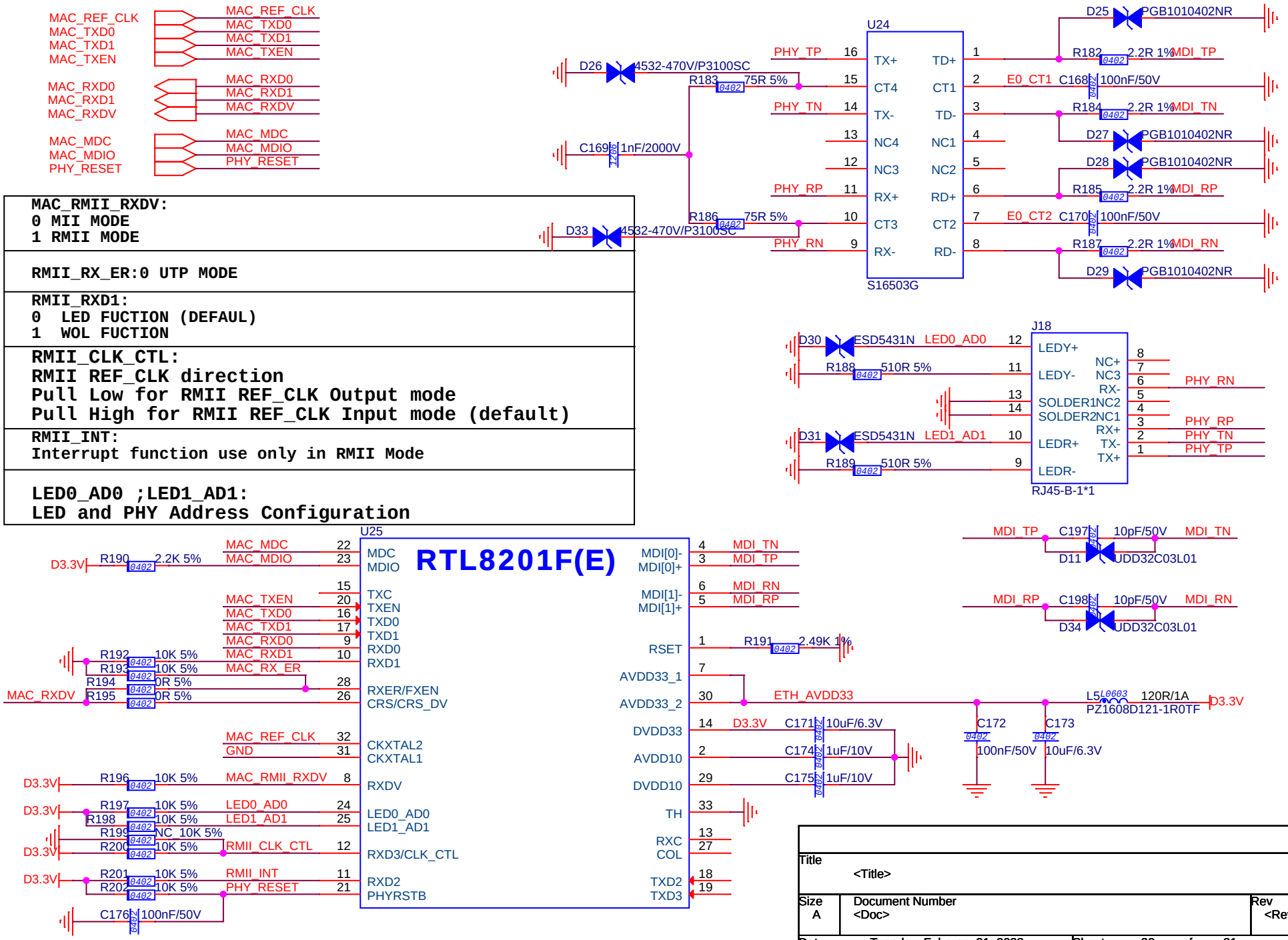
MAC_REF_CLK
MAC_TXD0
MAC_TXD1
MAC_TXEN
MAC_RXD0
MAC_RXD1
MAC_RXDV
MAC_MDC
MAC_MDIO
PHY_RESET
U24
S16503G
TX+
CT4
TX-
NC4
NC3
RX+
CT3
RX-
TD+
CT1
TD-
NC1
NC2
RD+
CT2
RD-
PHY_TP
PHY_TN
PHY_RP
PHY_RN
D26
4532-470V/P3100SC
D33
4532-470V/P3100SC
R183
75R 5%
R186
75R 5%
C169
1nF/2000V
E0_CT1
E0_CT2
C168
C170
100nF/50V
R182
R184
R185
R187
2.2R 1%
MDI_TP
MDI_TN
MDI_RP
MDI_RN
D25
D27
D28
D29
PGB1010402NR
MAC_RMII_RXDV:
0 MII MODE
1 RMII MODE
RMII_RX_ER:0 UTP MODE
RMII_RXD1:
0 LED FUCTION (DEFAUL)
1 WOL FUCTION
RMII_CLK_CTL:
RMII REF_CLK direction
Pull Low for RMII REF_CLK Output mode
Pull High for RMII REF_CLK Input mode (default)
RMII_INT:
Interrupt function use only in RMII Mode
LED0_AD0 ;LED1_AD1:
LED and PHY Address Configuration
J18
RJ45-B-1*1
D30
D31
ESD5431N
LED0_AD0
LED1_AD1
R188
R189
510R 5%
LEDY+
LEDY-
SOLDER1
SOLDER2
LEDR+
LEDR-
NC+
NC3
RX-
NC2
NC1
RX+
TX-
TX+
U25
RTL8201F(E)
MDC
MDIO
TXC
TXEN
TXD0
TXD1
RXD0
RXD1
RXER/FXEN
CRS/CRS_DV
CKXTAL2
CKXTAL1
RXDV
LED0_AD0
LED1_AD1
RXD3/CLK_CTL
RXD2
PHYRSTB
MDI[0]-
MDI[0]+
MDI[1]-
MDI[1]+
RSET
AVDD33_1
AVDD33_2
DVDD33
AVDD10
DVDD10
TH
RXC
COL
TXD2
TXD3
D3.3V
R190
2.2K 5%
MAC_MDC
MAC_MDIO
MAC_TXEN
MAC_TXD0
MAC_TXD1
MAC_RXD0
MAC_RXD1
MAC_RX_ER
MAC_RXDV
R192
R193
R194
R195
10K 5%
0R 5%
MAC_REF_CLK
GND
MAC_RMII_RXDV
R196
R197
R198
R199
R200
R201
R202
NC_10K 5%
RMII_CLK_CTL
RMII_INT
PHY_RESET
C176
100nF/50V
R191
2.49K 1%
ETH_AVDD33
C171
C174
C175
10uF/6.3V
1uF/10V
C172
C173
100nF/50V
10uF/6.3V
L5
PZ1608D121-1R0TF
120R/1A
C197
C198
10pF/50V
D11
D34
UDD32C03L01
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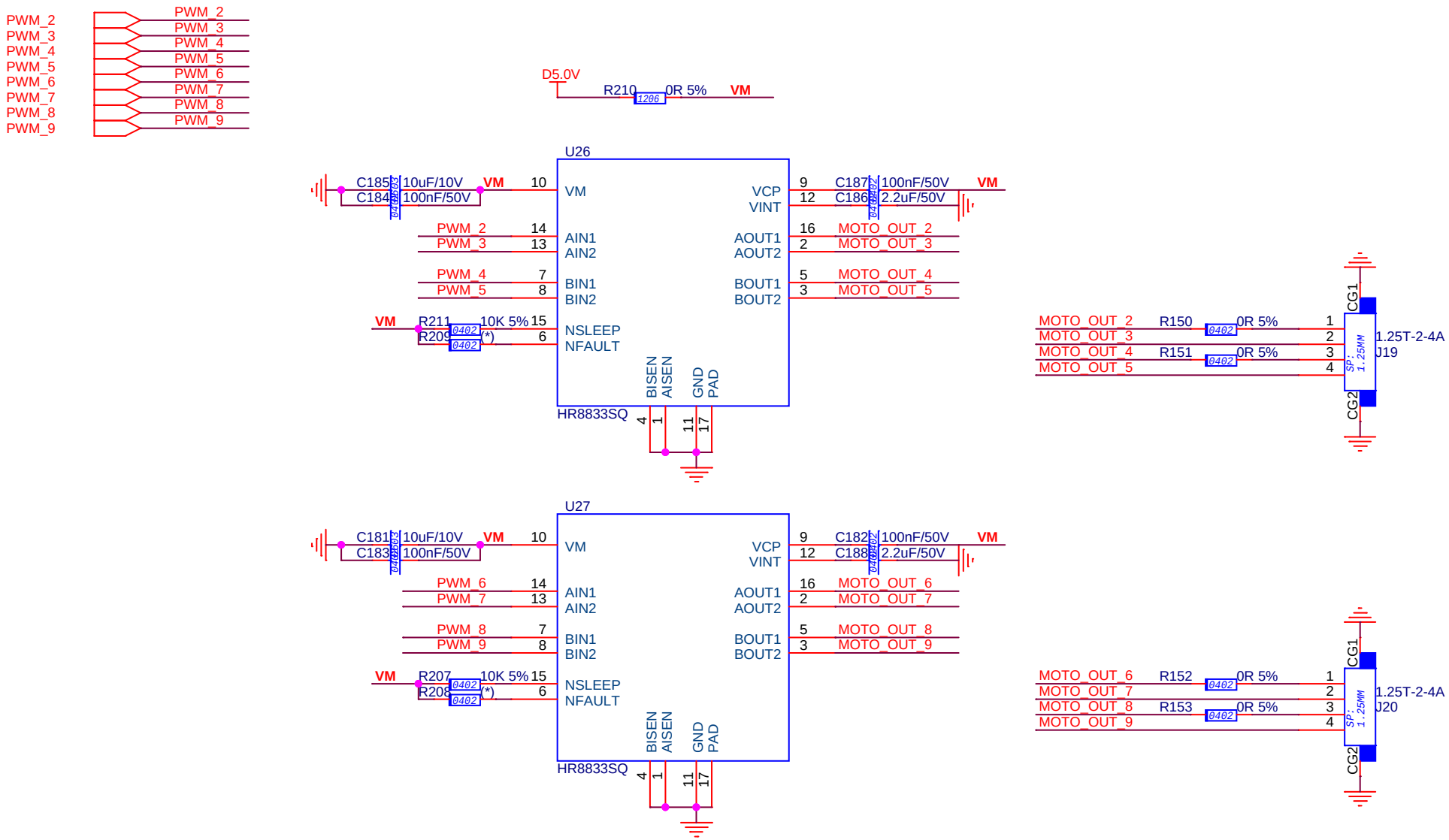

HR8833SQ / AT8833CQ / DRV8833RTY
QFN16_4*4mm
封装兼容

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